Agent based modelling

Social networks and opinion models

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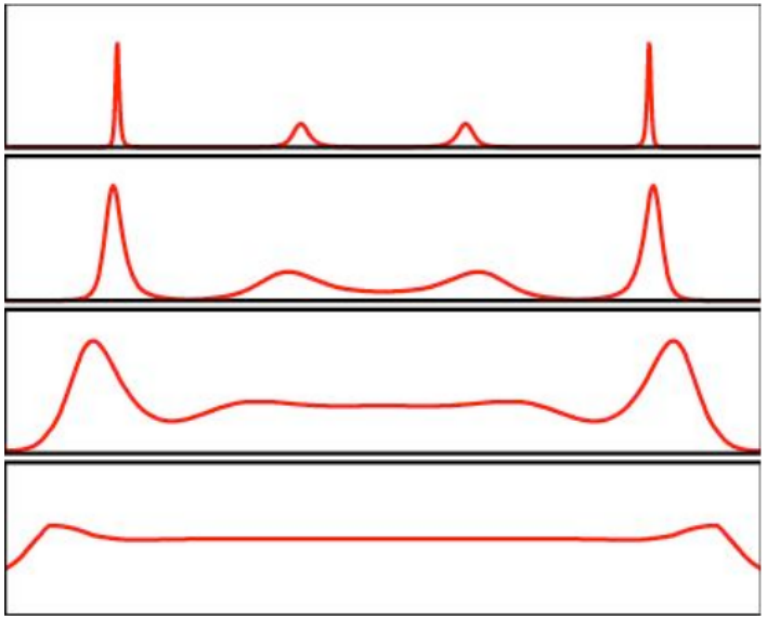
Bounded confidence opinion model: Deffuant model

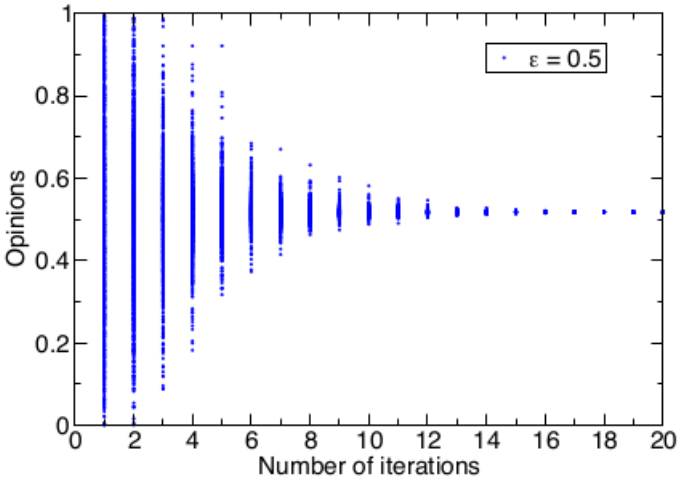
- Agents have opinion x_i
- if $|x_i(t) - x_j(t)| < \varepsilon$ then
  - $x_i(t+1) = x_i(t) - \mu[x_i(t) - x_j(t)]$
  - $x_j(t+1) = x_j(t) + \mu[x_i(t) - x_j(t)]$
- μ compromise parameter $\mu = 1/2$ complete compromise
- ε tolerance parameter
- Methods:
  - Monte-Carlo simulation
  - Master equation:

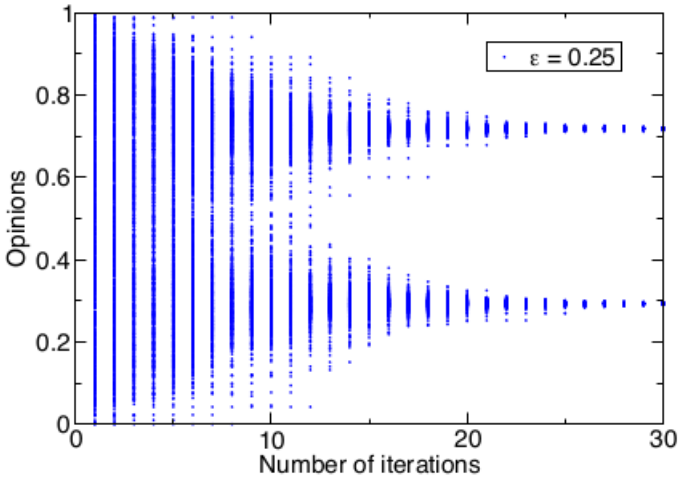
$$\frac{\partial P(x,t)}{\partial t} = \int_{|x_1 - x_2| < \varepsilon} dx_1 dx_2 P(x_1,t) P(x_2,t) \times$$

$$\times \left[\delta\left(x - \frac{x_1 + x_2}{2} \right) - \delta(x - x_1) \right]$$

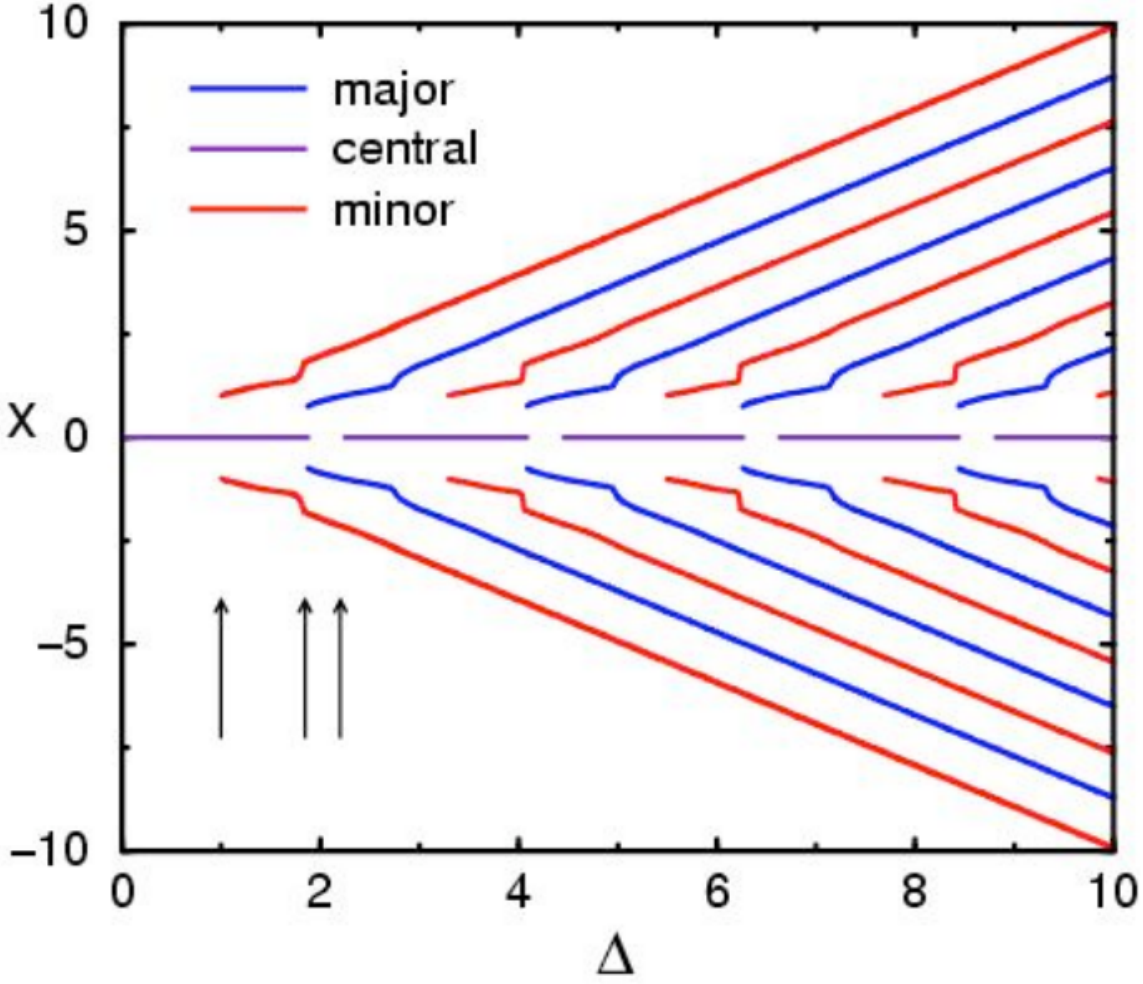
Deffuant model: Opinion groups (fully connected graph)







Deffuant model: Bifurcation diagram

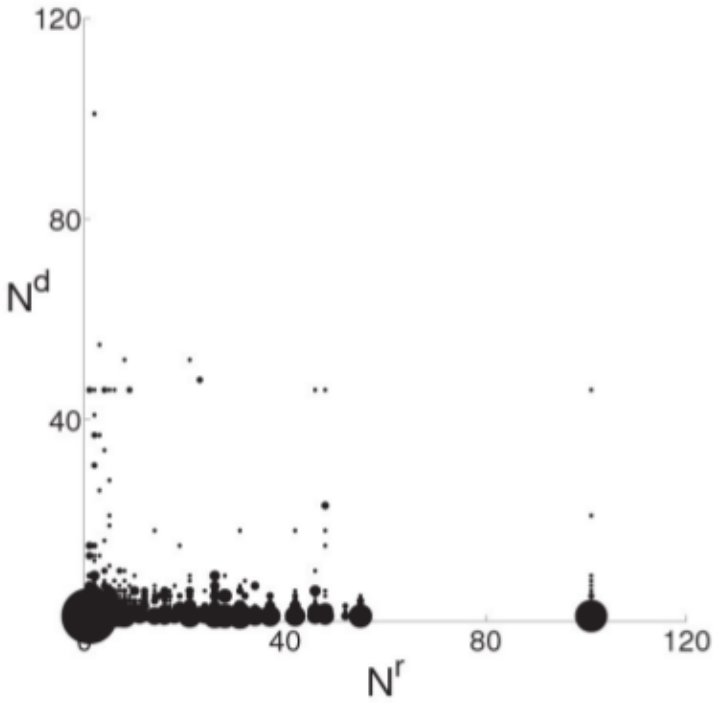


$\Delta = 2/\varepsilon$, $\mu = 1/2$

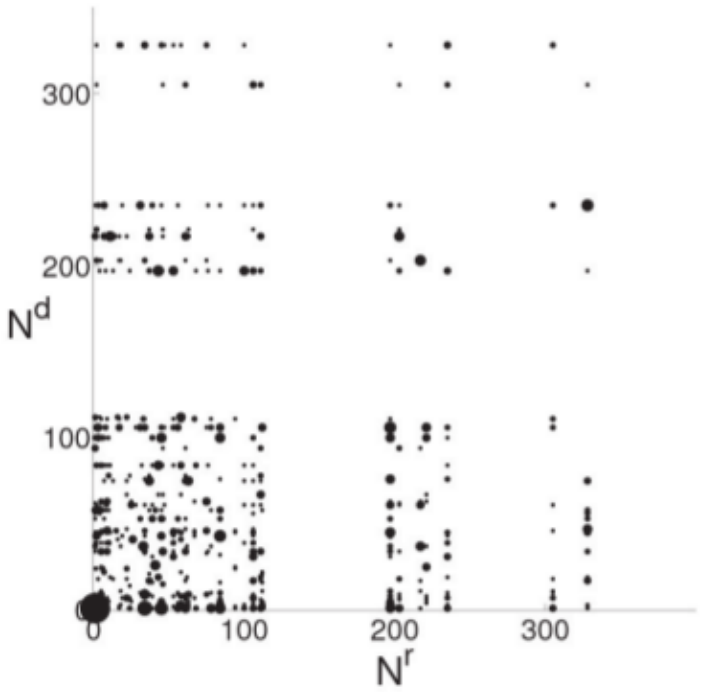
Application to Wikipedia edit wars

- Article has an opinion bieas A
- People who do not like the bias may edit the article
- Editors argue for a while then simply revert the other edits
- expeiremce of the editor helps to distinguish between edit war and vandalism

Benjamin Franklin



Israel and the apartheid analogy



Controversial words in article titles in 2009

people
War
Club
Israel
Iran
Arab
Ali
Mexico
Social
State
language
Country
United
Israeli
Jesus
Party
Islam
Muhammad
Lebanon
Jehovah
Basque
History
Race
Holocaust
Atlético
Czech
Madrid
Spain
France
Republic
Iranian
World
Jerusalem
Opus
Atheism
Union
Reza
Valencian
Bah
Chile
God
Adolf
Spanish
Global
Witnesses
Anarchism
Homeopathy
Gaza
Iraq
Fox
Jews
Manuel
Yitzhak
Pseudoscience
Gulf
Ball
Big
Islands
theories
Dragon
Church
Spears
Messi
Argentine
Liberalism
Democracy
Best
National
Civil
Death
Cuba
Persian
Arabia
Cristiano
Martin
Kosovo
Pavel
Northern
Kurdish
Ahmad
Joseph
Simpsons
Evolution
Sexual
White
Felipe
Mir
Kook
Mousavi
Christianity
Revolution
Hitler
Catholic
Rabin
Pinochet
Syria
Hapoel
John
Ronaldo
Five
Politics
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Velez
Truth
Tehran
Central
Aviv
Chemical
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Acropolis
Joaquin
XVI
Lionel
Pamplona
Community
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Ashley
airline
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Andr s
Ariel
Libertarianism
Hezbollah
Vienna
Bilbao
Madrid
Football
Fútbol
Antena
Workers
Primera
Triumph
Almagro
train
Animal
Mohammad
City
Naruto
Mahmoud
Vladimir
Romance
Fernández
Abortion
Belinda
Croatia
Religion
Cristina
Federer
Karl
Canary
Peru
bombing
Egypt
Haifa
Michael
Kahane
Panda
Saudi
Laden
Real
Southern
Avellaneda
Rosario
Britney
Shiraz
David
Alianza
Pahlavi
Dion
Ron
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Jiménez
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Václav
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Roland
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Tunica
Muslim
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Venezuela
Augusto
Bogotá
Hussein
Casablanca
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Astrology
Wales
Daniel
Progress
Morocco
Banu
Nicolaus
Barack
Calder n
Michelle
Wikipedia
Saint
Francisco
Ehud
Deportivo
Medell n
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conspiracy
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Iberia
Guayaquil
Manos
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Libertarianism
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Pamplona
Award
Freemasonry
warming
Joseph
Simpsons
Valencian
Reza
Union
Jerusalem
Northern
Kurdish
Acropolis
Ahmad
Tamim
Kosovo
Pavel
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movement
Chemical
Hurac n
Independiente
Obama
Google
Aviv
election
Kook
Homosexuality
Palestinian
entertainer
Latin
Composición
Descripción
Felipe
Mir
Central
Tehran
Christianity
Mousavi
Genocide
Baruch
Toluca
Lavín
Truth
Velez
Revolution
Iberia
Guayaquil
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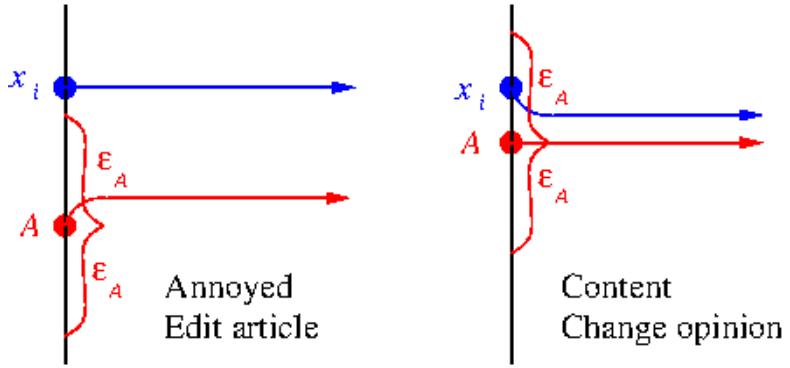
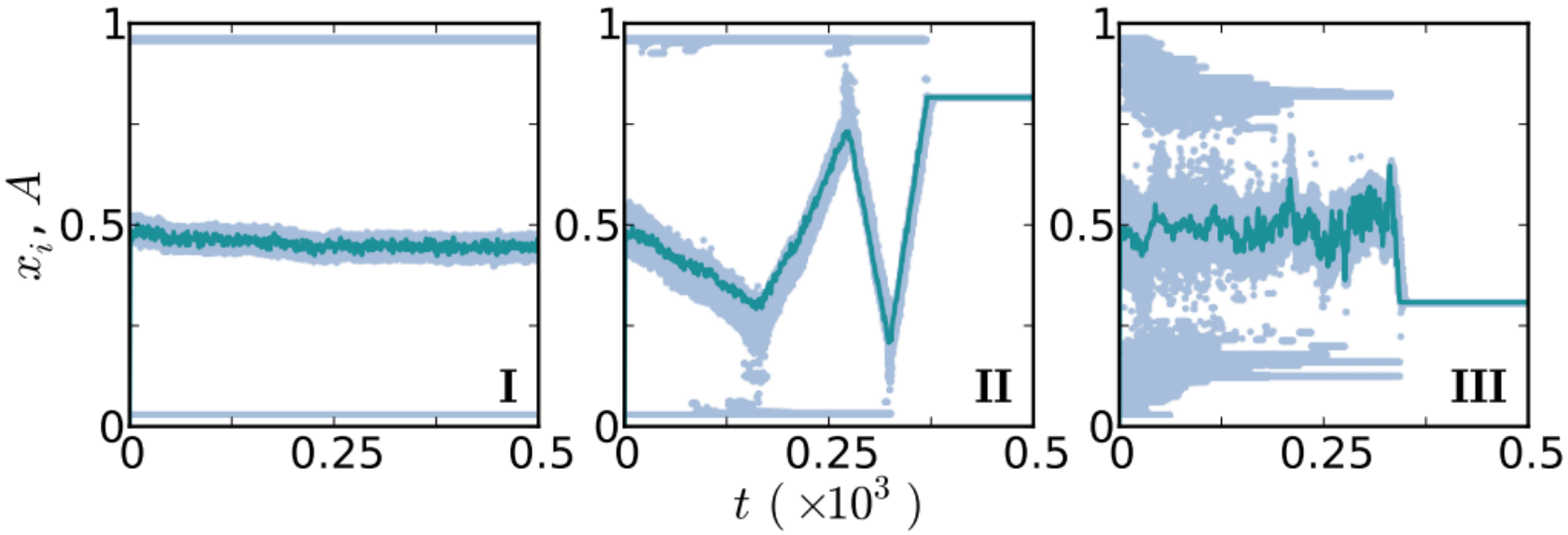
Composición
Descripción
Lionel
XVI
Joaquin
Andalucía
América

Bounded confidence model for Wikipedia

- Agents have opinion x_i, Article A
- if $|x_i(t) - A(t)| < \varepsilon_A$ then
  - $x_i(t+1) = x_i(t) + \mu_A[A(t) - x_i(t)]$
- Article bias is close to the editor's no edit but agent is a bit influenced by the article
- if $|x_i(t) - A(t)| < \varepsilon_A$ then
  - Article bias is far from the editor's edit!
  - $A(t+1) = A(t) + \mu_A[x_i(t) - A(t)]$
- μ_A article weight (probably amount of content or reasoning)
- ε_A tolerance parameter



Results of the Wikipedia model



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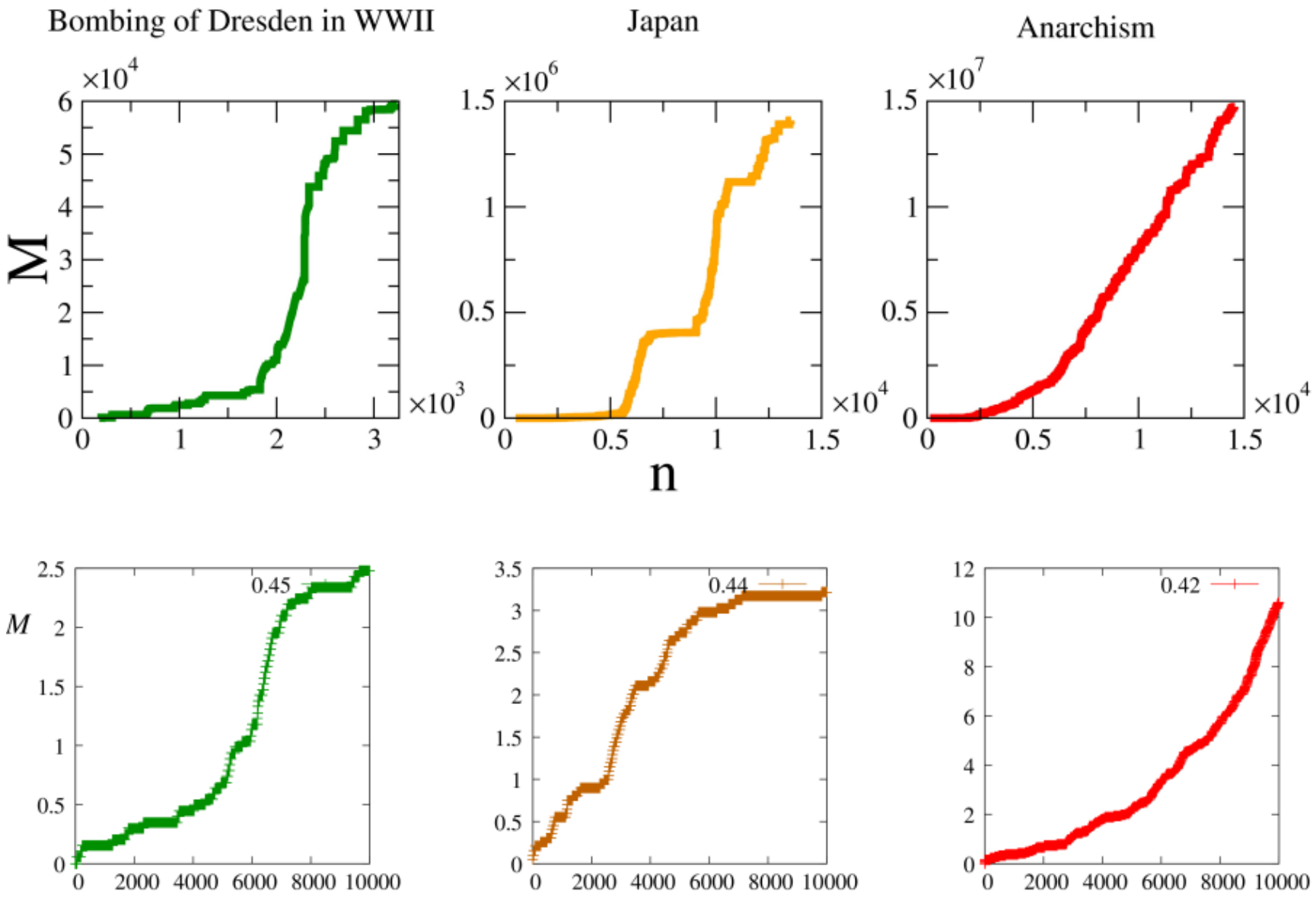
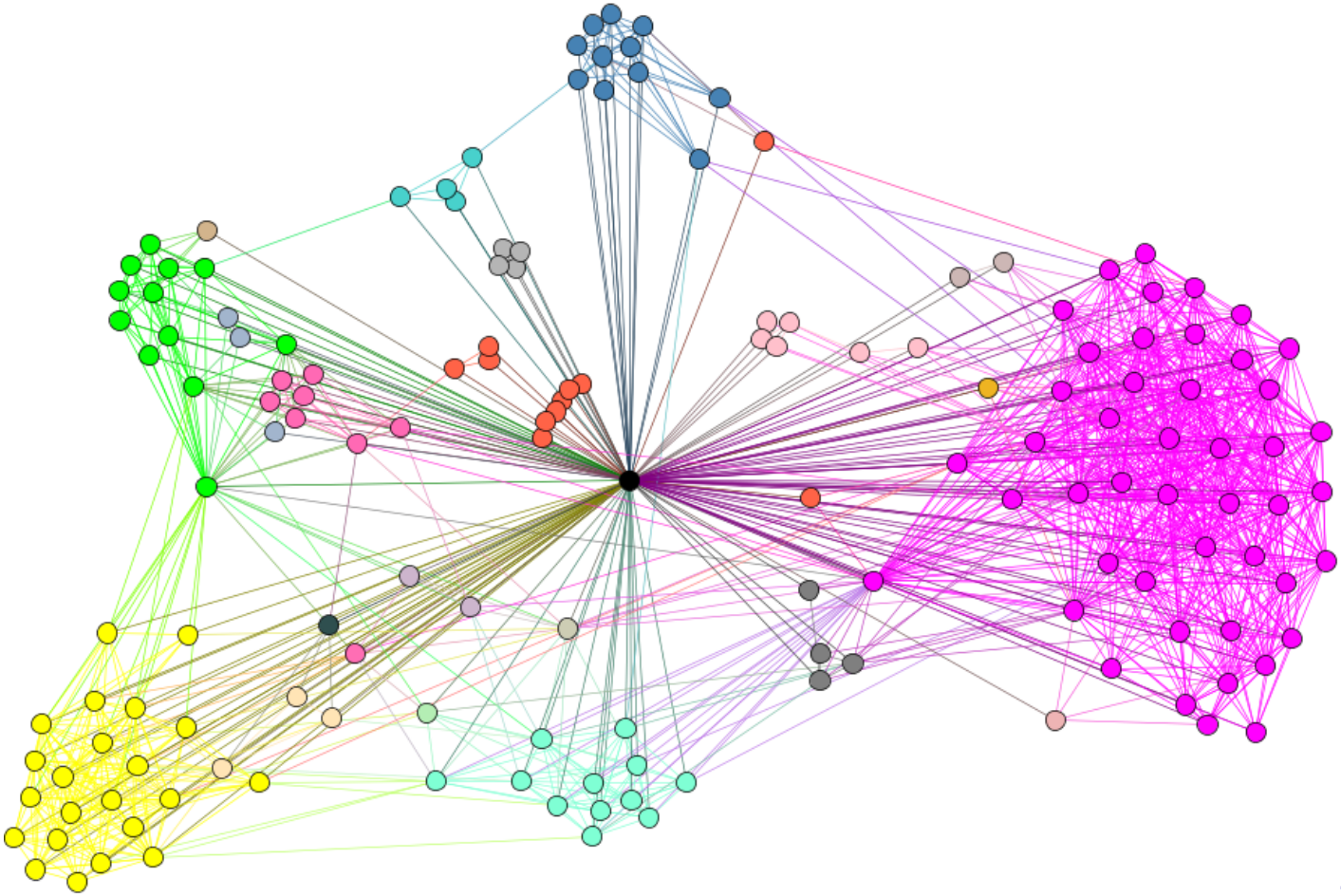


Growing networks

- Simulate real life
- Use minimal elements
- Do not incorporate effect what one wants to recover
- Example: simulate social network (modular)



Growth models

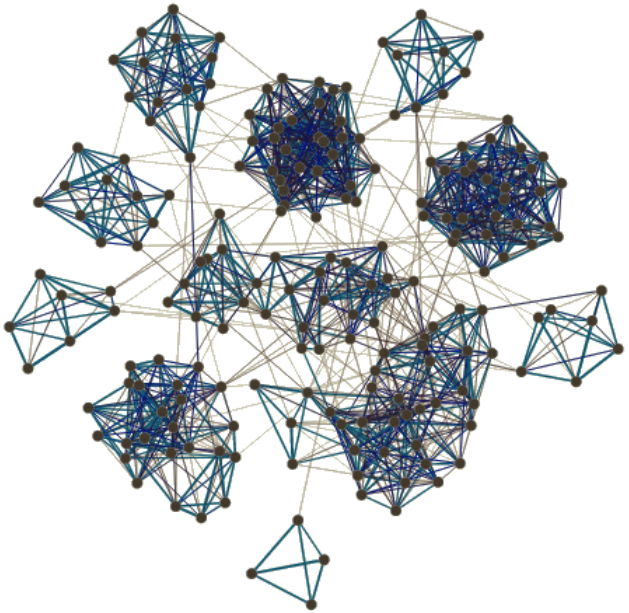
- Barabási-Albert model: Simple growth mechanism, preferential attachment, model for Internet
- More complicated systems?
- Two version of a simple model for social networks

Social networks

- Human relation
- Very complicated dynamics
- Not really a growth model, more a dynamics steady state
- Observations:
  - Weighted network
  - Large clustering coefficient (friend of friends usually know each other)
  - Not scale free
  - Small world
  - Granovetter: Strength of the weak ties

Granovetter: Strength of the weak ties

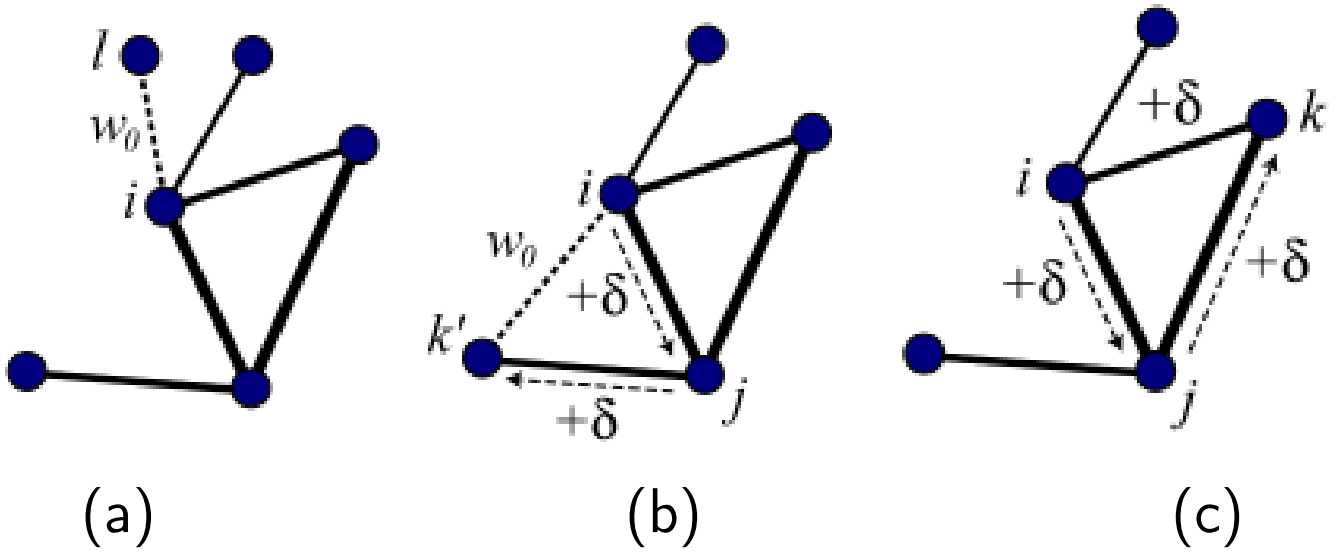
- Human groups are strongly connected
- There are weak connections connecting the groups
- These weak connections mean sproadic meeting
- Important for information flow
- Example: Find a job



Granovetter, Mark S. "The strength of weak ties." American journal of sociology 78.6 (1973): 1360-1380.

Kumpula model

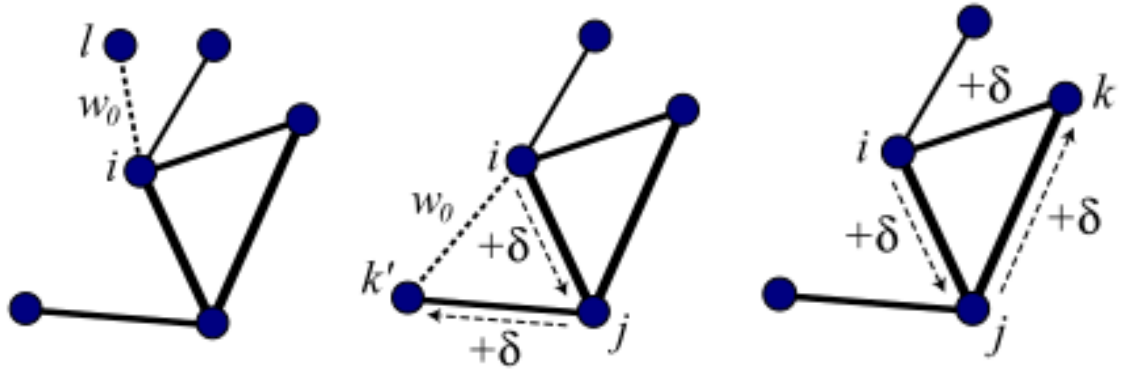
- N nodes (originally unconnected)
- (a) Randomly meet someone (low probability) global attachment
- (b) Two friends of someone get to know each other, cyclic closure
- (c) An already present triangle gets strengthened



JM. Kumpula, Emergence of communities in weighted networks. PRL, 99(22), 228701 (2007).

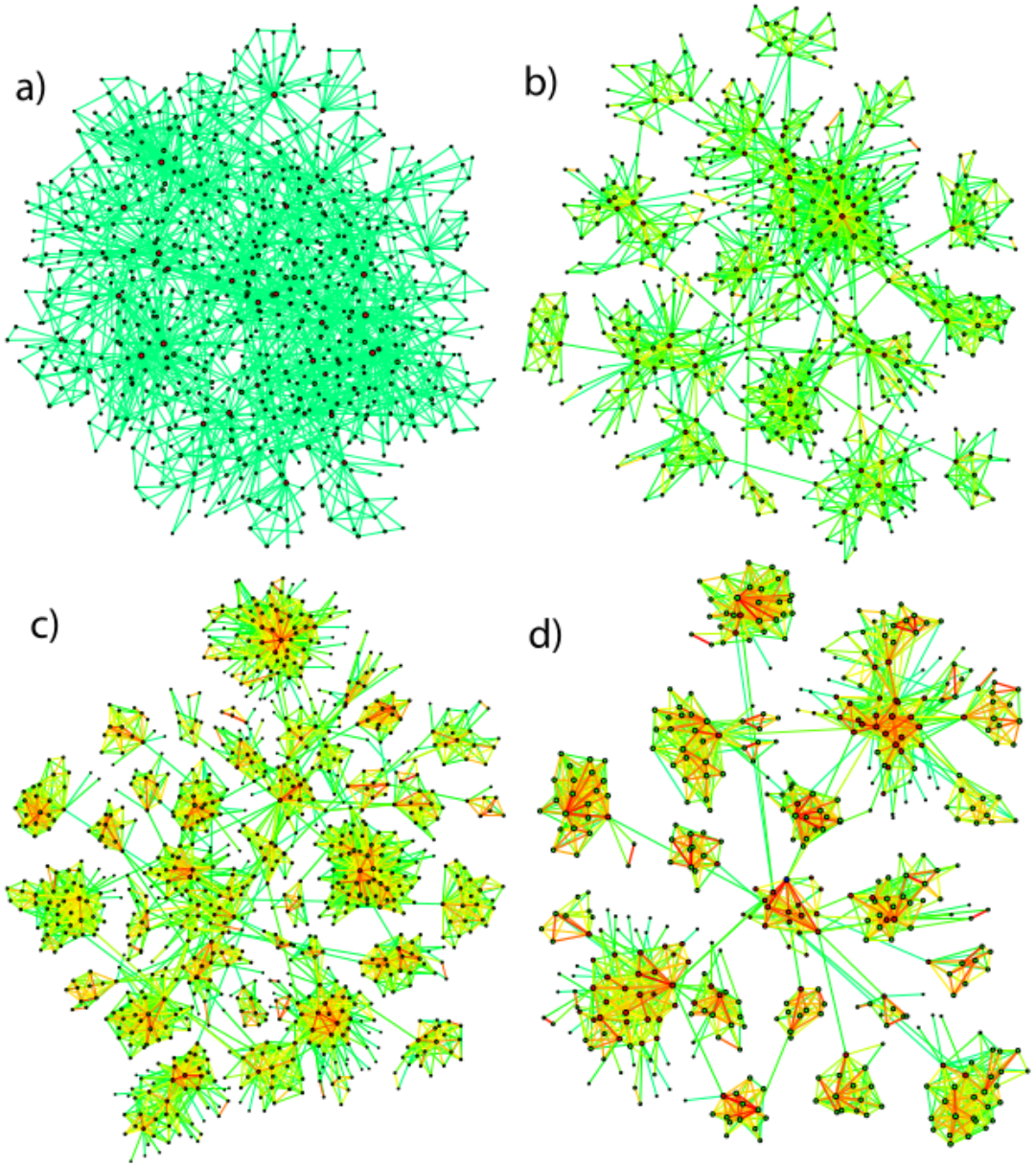
Kumpula model

- N nodes (originally unconnected)
- (a) (with prob. p_r) random link to an unconnected node. Link weight w_0
- (with prob. p_d) i selects friend j with prob. proportional to the link weight. j selects friend k similarly. Both links are strengthened by δ. Two cases:
  - (b) There is no link between i and k: create a link with p_Δ with weight w_0
  - (c) There is a link between i and k: strengthen by δ
- (d) (with prob. p_d) clear the links of a node (enforce steady state, there are more realistic versions)

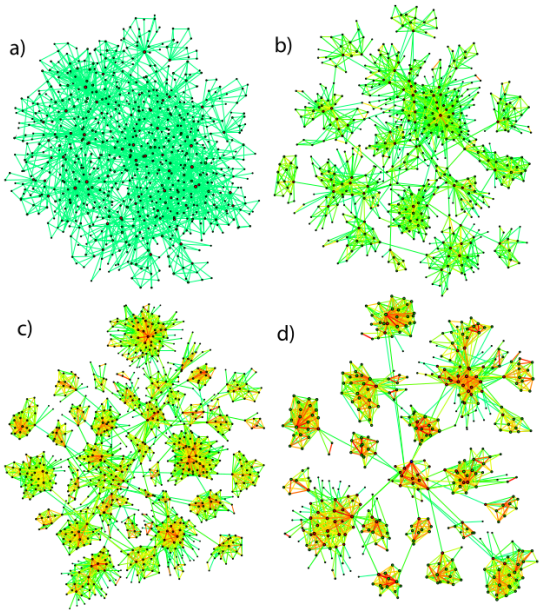


(a) (b) (c)

Kumpula model: results ($\delta = 0,\ 0.1,\ 0.5,\ 1.0$)



Kumpula model: results



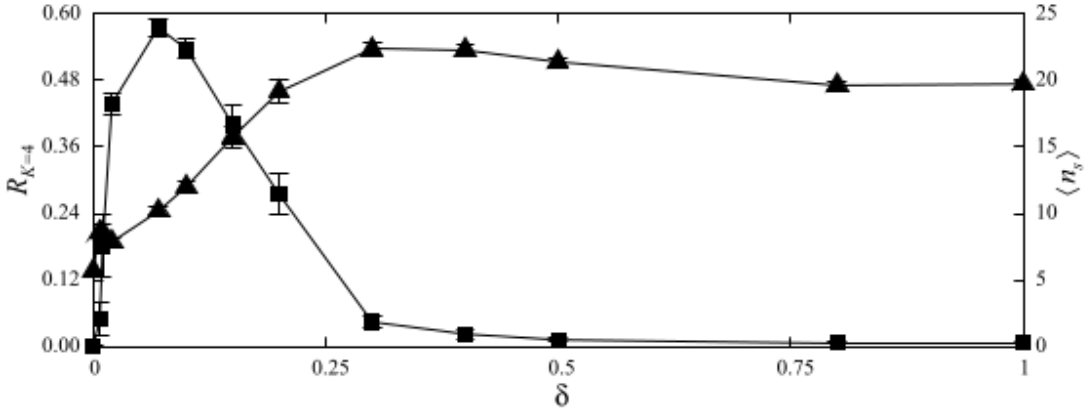


FIG. 3: $R_{k=4}$ (□) and $\langle n_s \rangle$ (△) as a function of δ. Results are averaged over 10 realizations of $N = 5 \times 10^4$ networks. Error bars are measured standard deviations.

Kumpula model: results

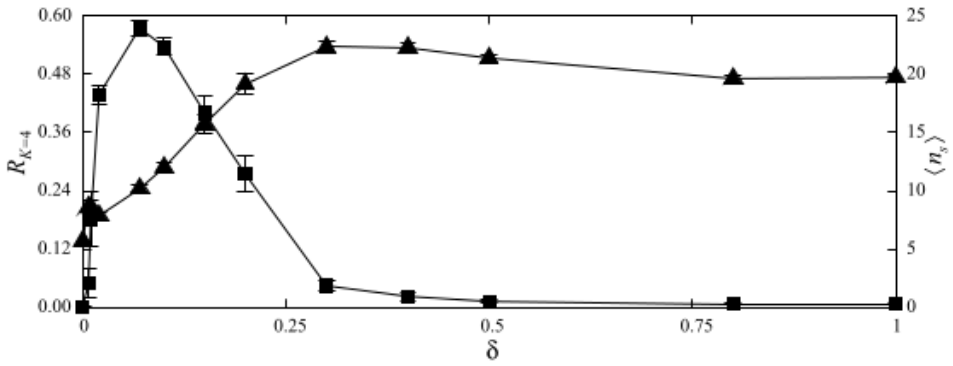
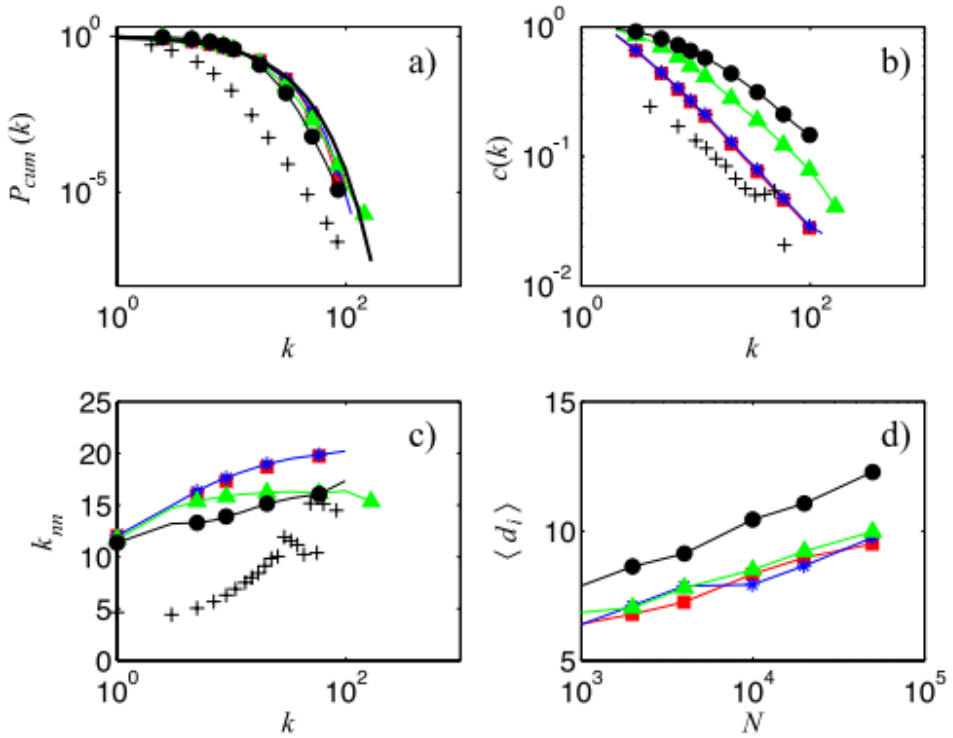


FIG. 3: $R_{k=4}$ (□) and $\langle n_s \rangle$ (△) as a function of δ. Results are averaged over 10 realizations of $N = 5 \times 10^4$ networks. Error bars are measured standard deviations.



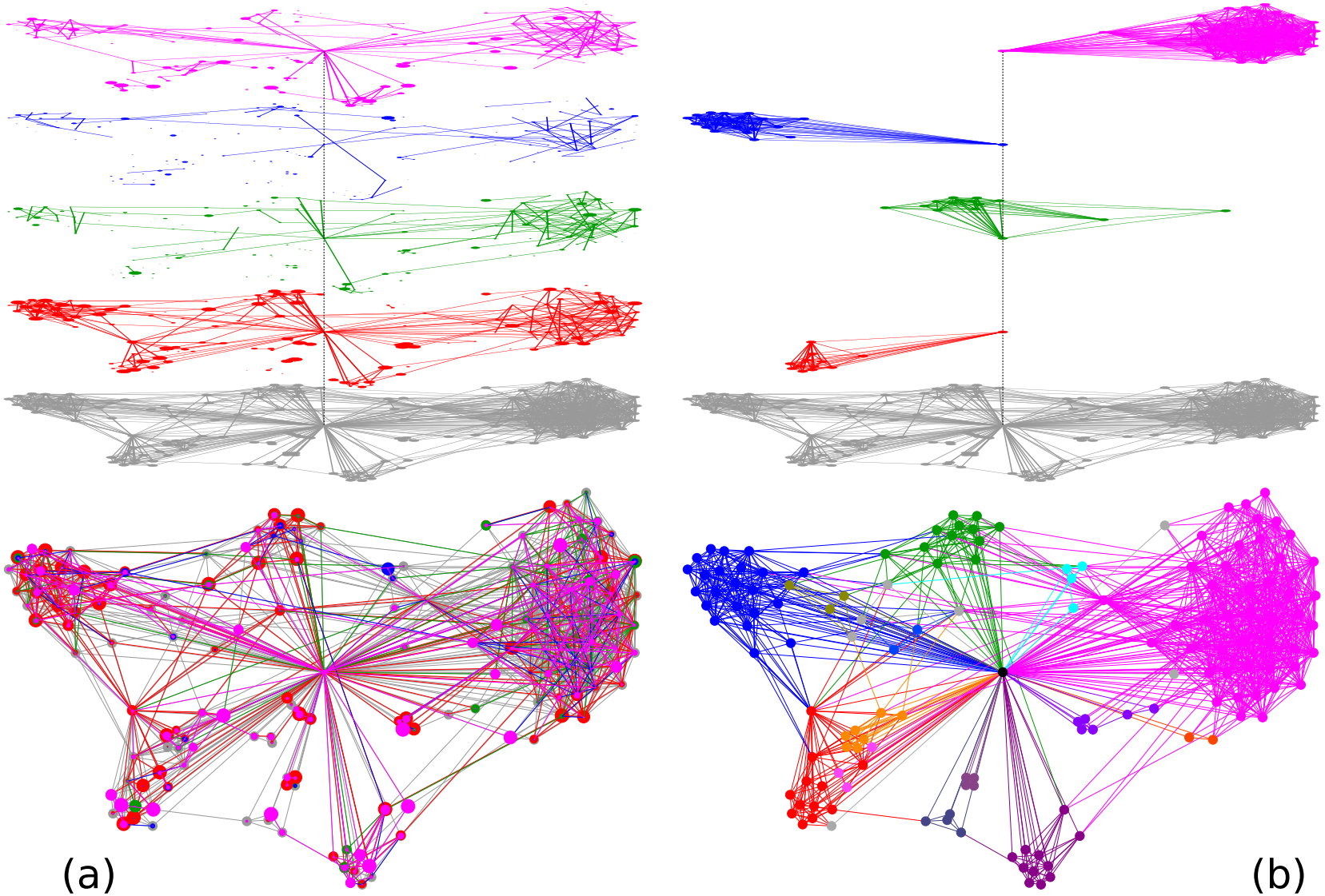
Kumpula model: results

- Very simple assumptions
- Emergence of community structure (depending on parameters)
- Good to test effects of elementary processes on global structure
- Not apt for recovering well defined structures

Multiplex networks: Social networks

Communication channel

Social context



Axelrod model of dissemination of culture

- Each individual is endowed with a certain culture
- They have cultural needs and preferencies therein
- An individual's culture is characterised by a list of F features
- Each feature has q different traits
- Assumptions
  - people are more likely to interact with others who share many of their cultural attributes
  - these interactions tend to increase the number of cultural attributes they share (thus making them more likely to interact again).

Axelrod 1997

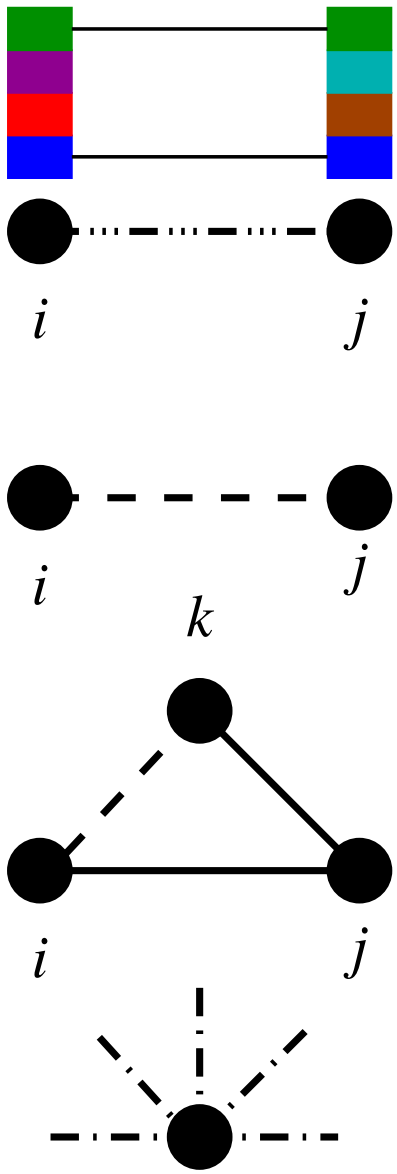
Axelrod model of dissemination of culture

- Model
  - One agent k (active) is selected at random.
  - One of agent k's neighbours, denoted agent r (passive), is selected at random.
  - n_{kr} number of features in which agents k and r matches
  - Agents k and r interact with probability equal to their cultural similarity n_{kr}/f
  - The interaction consist of k copying one of the unmatched features of agent r
  - In this way, agent k approaches agent r's cultural interests

MOVIE

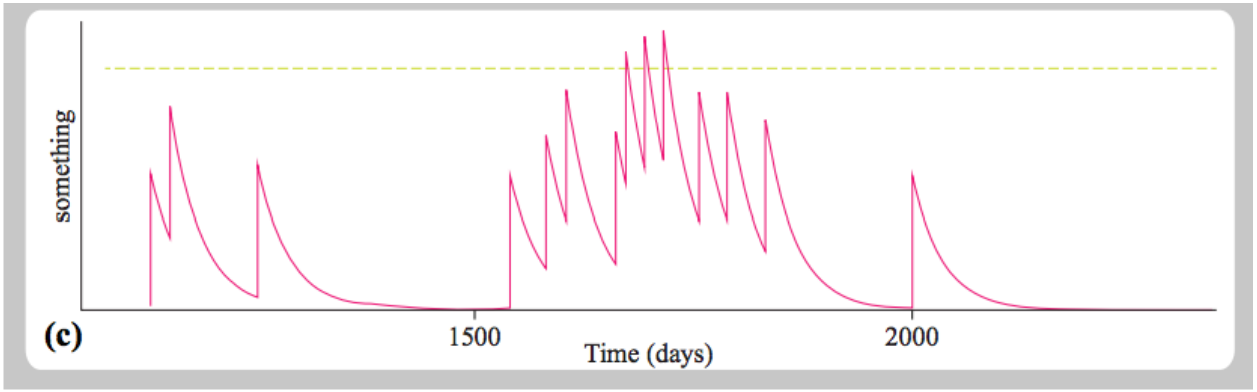
Multi layer model of social networks

- Peaple have F social features with q values each
- Ego first selects feature (s)he wants to do some social action
- (S)he can do it only with people with matching the specific feature
- Random connection, rare
- Triangles: common
  - Link selection proportional to weight
  - Link establishment with some probability and strengthening participating links
  - Link aging

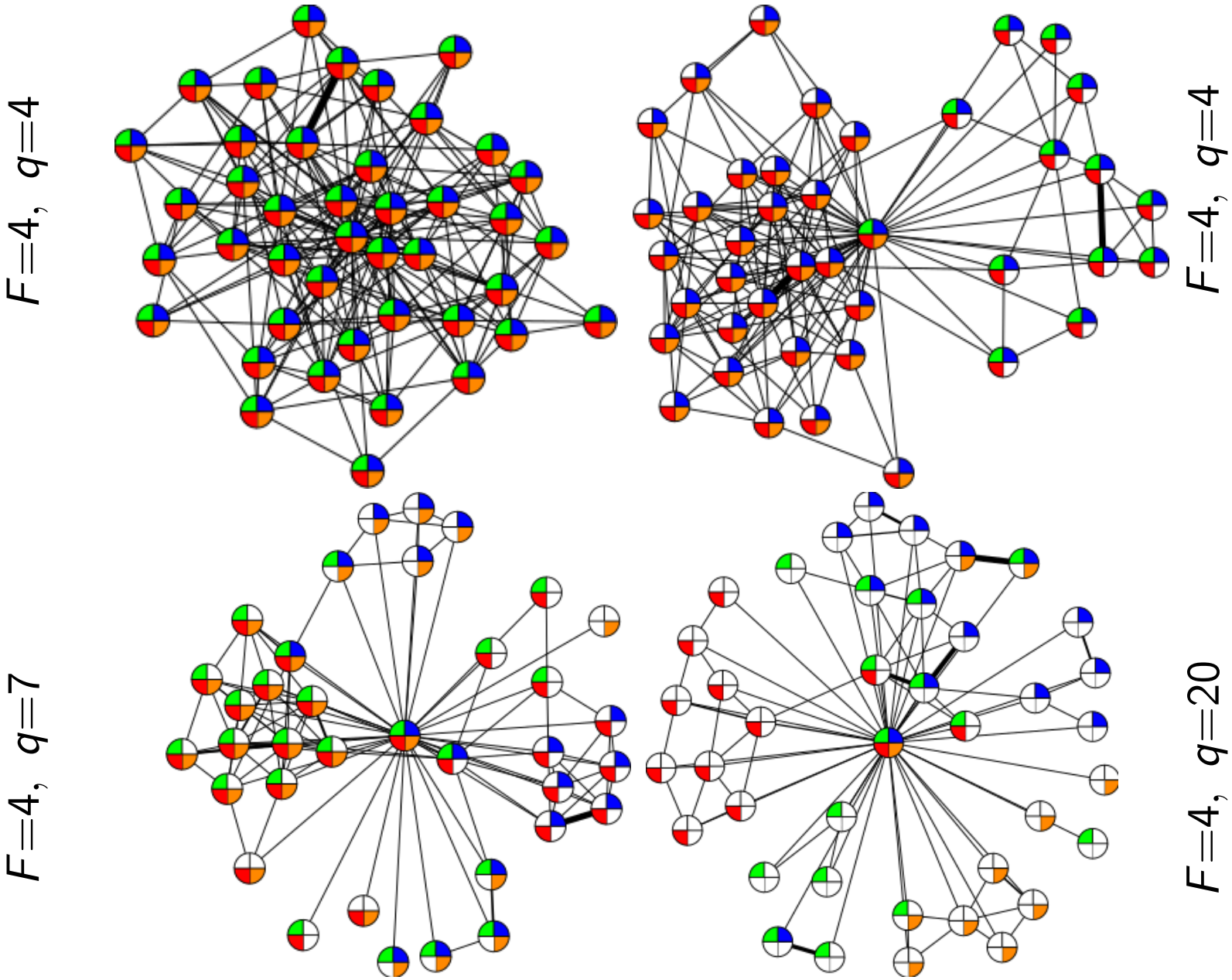


Link aging

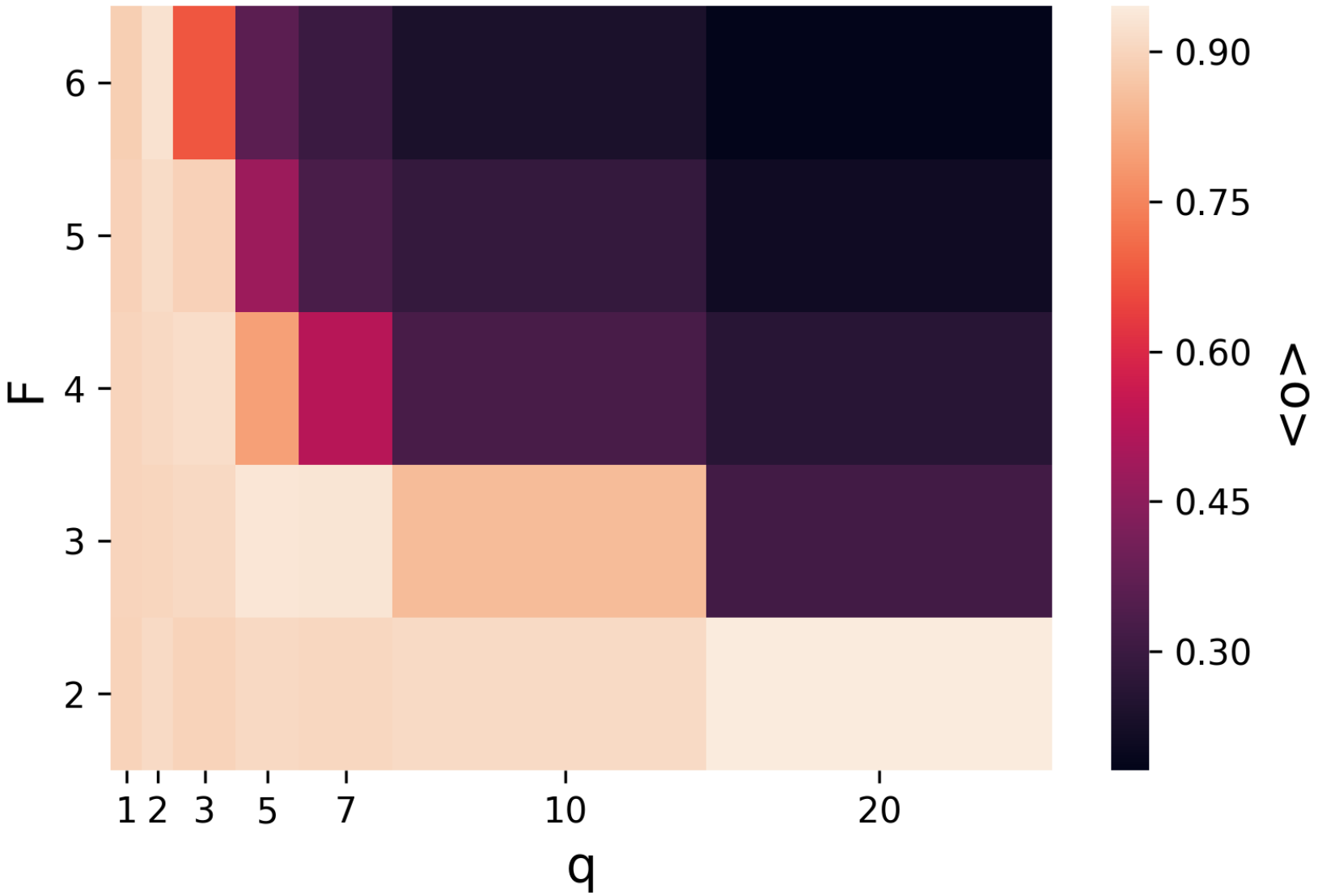
- Steady state
- Relationships fade with time
- Communication is an instantaneous strengthening



Multilayer social model: egocentric networks



Multilayer social model: Phase diagram



Multilayer social model: Conclusions

- Possible explanation for echo chambers:
  - People want a well determined number of friends around them
  - People look for people with whom they share at least some idea/activity
  - Due to the inferent strengthening mechanism with those with whom we are more similar we end up haveing a set of friends as similar as possible
  - If the number of important features or the number of options are small we can fill up our friendship circles with friends thinking the same way as we do